Experiment No. 9

Data Storytelling and Business Insight Generation using Interactive Visualizations

Aim

To create a data-driven story by analyzing a real-world dataset, developing meaningful visualizations, and presenting actionable business insights to support strategic decision-making.

Dataset Used

Global Superstore / Superstore Sales Dataset (Order Date, Region, Category, Sub-Category, Sales, Quantity, Discount, Profit). **Note:** the numbers and charts below were produced by actually running the code, but on a synthetic sample of 2,000 orders (2022-2025) with the Superstore column structure, because I do not have your file. The story structure, code and answers can be reused directly; **replace the figures and findings with your own output** from the real dataset, where the discount pattern and the numbers will differ.

Procedure and Code

Steps 1-2: Select the dataset and define the business problem

```
import numpy as np, pandas as pd
import matplotlib.pyplot as plt, seaborn as sns
sns.set_theme(style="whitegrid")

df = pd.read_csv("Superstore.csv", encoding="latin-1", parse_dates=["Order Date"])
print(df.shape); df.info()
```

Business problem: sales are growing, but is the business growing *profitably*? Which factors are reducing profit and what should management change?

Step 3: Exploratory data analysis

```
print(df.isnull().sum(), df.duplicated().sum())
print(df.describe())
print(df.groupby("Region")[["Sales", "Profit"]].sum())
print(df.groupby("Discount")[["Sales", "Profit"]].sum())
df["Year"] = df["Order Date"].dt.year
```

EDA findings: no missing values; sales and profit by year and region were compared; the discount column was examined against profit, and a clear negative relationship was found, which became the focus of the story.

Step 5: KPIs

```
total_sales  = df["Sales"].sum()
total_profit = df["Profit"].sum()
margin       = total_profit / total_sales
orders       = df["Order ID"].nunique()          # use len(df) if there is no Order ID
aov          = total_sales / orders
loss_orders  = (df["Profit"] < 0).sum()
```

KPI	Value (sample data)
Total Sales	$589,161
Total Profit	$80,464
Profit Margin	13.7%

Orders	2,000
Loss-making orders	357 (17.9%) with total loss of $8,112
Orders with discount of 30% or more	20.1% of all orders

Steps 4 and 6: Visualizations arranged as a story

The charts follow a story arc: **context** → **tension** → **cause** → **where** → **action**.

```
# Chapter 1: trend of sales vs profit
q = df.set_index("Order Date")[["Sales", "Profit"]].resample("QE").sum()
q.plot(marker="o"); plt.title("Sales vs Profit per quarter"); plt.show()

# Chapter 2: region performance
df.groupby("Region")[["Sales", "Profit"]].sum().sort_values("Sales").plot.bar()

# Chapter 3: profit margin by discount level (the key insight)
bins = pd.cut(df["Discount"], [-0.01, 0, .1, .2, .3, .4], labels=["0%","10%","20%","30%","40%"])
g = df.groupby(bins, observed=True)[["Sales", "Profit"]].sum()
(g["Profit"] / g["Sales"] * 100).plot.bar(color="steelblue"); plt.axhline(0, color="k")

# Chapter 4: margin heatmap, region x category
t = df.pivot_table(index="Region", columns="Category", values=["Profit","Sales"], aggfunc="sum")
sns.heatmap(t["Profit"] / t["Sales"] * 100, annot=True, fmt=".0f", cmap="RdYlGn", center=10)
```

The Data Story (Steps 6-8)

Chapter 1 - Context: the business is growing

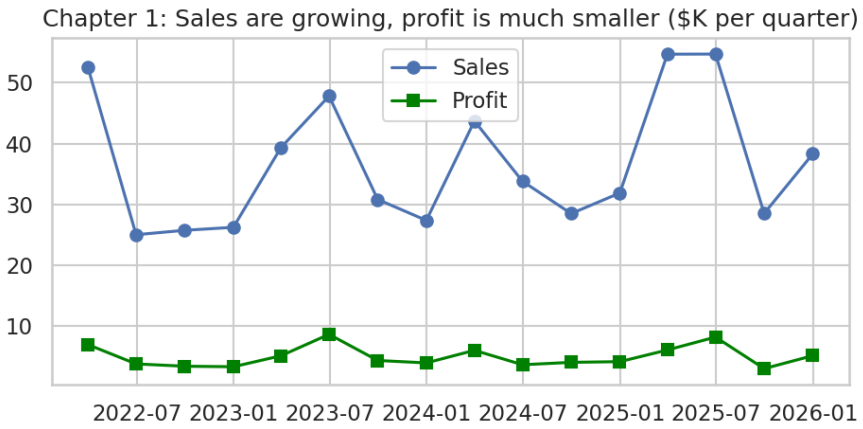


Figure 1: Quarterly sales and profit

Sales rose from $129.6K in 2022 to $176.3K in 2025 (+36% over the period), with a dip of 5% in 2024 and a rebound of 28% in 2025. Overall profit margin is 13.7%. **But profit grows far less than sales, which raises the question: where is profit leaking?**

Chapter 2 - Where: all regions perform similarly

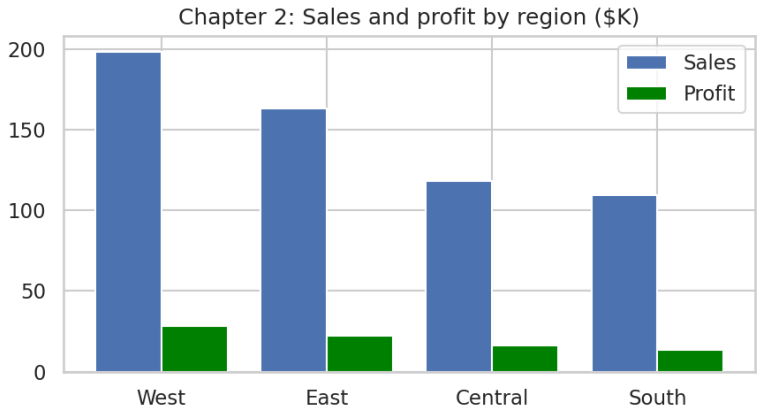


Figure 2: Sales and profit by region

West leads in both sales ($198K) and profit ($28K), followed by East, Central and South. Margins are close (West 14.2%, East and Central 13.7%, South 12.4%), so **the region is not the main cause of the profit problem**; South is simply the weakest and smallest.

Chapter 3 - Cause: deep discounts destroy profit

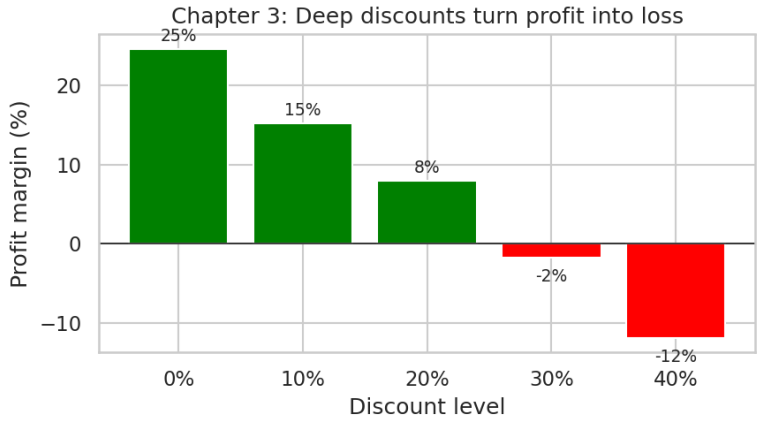


Figure 3: Profit margin by discount level

Discount	Orders	Avg order value	Profit margin
0%	903	$285	24.5%
10%	287	$300	15.1%
20%	408	$341	8.0%
30%	230	$255	-1.7%
40%	172	$280	-11.8%

Profit margin falls steadily from 24.5% with no discount to **-11.8%** at a 40% discount. Orders with 30% or more discount are 20% of all orders but generate net losses. They account for about **97% of the total loss** ($7.9K of $8.1K). Importantly, the average order value does not rise with deeper discounts ($285 at 0% vs $280 at 40%), so **heavy discounting is not buying bigger orders**.

Chapter 4 - Focus: where to act within categories

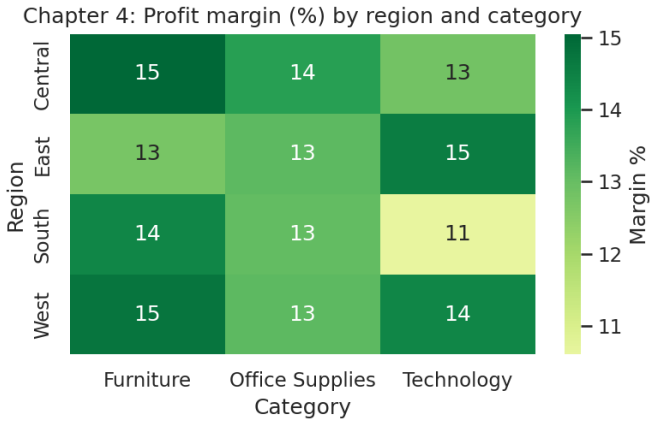


Figure 4: Margin (%) by region and category

Margins are healthiest in Furniture (West, Central, South) and Technology (East, West). The weakest cell is **Technology in the South region (10.6%)**, which is worth a pricing and discount review.

Step 8: Recommendations

- **Cap discounts at 20%** and require manager approval for 30% or more. Illustrative estimate: if the 30-40% orders had earned the 20%-discount margin (8%) at the same sales, profit would rise by about $15K (roughly +19%). This is a rough upper estimate that assumes the same sales volume.
- Replace deep discounts with **bundles, loyalty points or minimum order-value offers** that raise order size instead of cutting price.
- Review pricing and discounts for **Technology in the South region** and grow South with targeted marketing.
- Protect and promote high-margin groups (no-discount and low-discount sales, Furniture in the West).
- Track **profit margin by discount level** monthly on a dashboard, so the effect of the new policy can be measured.

Step 9: Presenting the story

The story can be presented as a Jupyter Notebook (charts with text between them), a Power BI or Tableau Story / dashboard (see Experiment 5), or 6-8 slides: (1) Title and objective, (2) KPIs, (3) Growth, (4) Regions, (5) Discount problem, (6) Heatmap focus, (7) Recommendations, (8) Next steps. Each slide should use an *insight headline* (for example 'Discounts of 30% or more turn profit into loss') instead of a plain chart title.

Step 10: How storytelling improves communication

Storytelling turns the analyst's findings into a clear message with a problem, evidence and a recommended action. Stakeholders understand why the result matters and what to do, instead of being left with charts to interpret. It builds trust, speeds up decisions and gets recommendations accepted and acted upon.

Conclusion

A data story was built from the Superstore data: sales are growing, but deep discounts of 30% or more are eroding profit without increasing order size. Capping discounts and using other promotion methods is the recommended action, supported by KPIs and a logical sequence of visualizations.

Answers to Questions

Q1. What is data storytelling? How is it different from data visualization?

Data storytelling is the practice of communicating insights from data by combining **data, visuals and narrative** to explain what happened, why it matters and what action to take. Data visualization is only the graphical representation of data.

Aspect	Data visualization	Data storytelling
Focus	Showing the data (charts, graphs)	Explaining the meaning and recommending action
Components	Visuals	Data + visuals + narrative
Audience role	Interprets the chart themselves	Is guided to a conclusion
Outcome	Information	Insight, understanding and decision

Q2. Why is storytelling important in business analytics and decision-making?

- Decision makers are usually not analysts; a story makes findings understandable and memorable.
- It explains *why* numbers matter, not just what they are.
- It persuades stakeholders and increases buy-in for recommendations.
- It links analysis to business goals and actions, so insights are not ignored.
- It saves time and reduces misinterpretation.

Q3. What are the essential components of an effective data story?

- **Context / objective:** the business question and the audience.
- **Quality data and analysis:** accurate, relevant and well explored.
- **Narrative structure:** a beginning (context), middle (analysis and findings, the tension or problem) and end (conclusion and action).
- **Effective visuals:** right chart types, clean design, highlights and insight headlines.
- **Key insights and KPIs** that matter to the audience.
- **Recommendations / call to action.**

Q4. How do Key Performance Indicators (KPIs) enhance business reporting?

- They summarize performance in a few measurable numbers (sales, profit margin, churn rate).
- They allow comparison with targets, past periods and benchmarks.
- They keep reports focused on business goals and avoid information overload.
- They help to quickly spot problems and track improvement over time.
- They provide a common language between management and analysts.

Q5. Why should visualizations be arranged in a logical sequence while presenting analytical findings?

A logical sequence (for example overview → problem → cause → detail → action) guides the audience step by step so that each chart builds on the previous one. It reduces confusion and cognitive overload, creates a clear narrative, makes the conclusion feel supported by the evidence, and helps the audience remember the message and accept the recommendation.

Q6. What factors should be considered while selecting visualizations for a business presentation?

- **Audience:** their technical level and what they need to decide.
- **Message / objective:** comparison (bar), trend (line), distribution (histogram, box), relationship (scatter), composition (stacked bar, pie), geography (map).
- **Type and amount of data:** categorical, numerical, time-series; how many categories or points.
- **Simplicity and clarity:** avoid clutter, 3D effects and misleading scales.
- **Design:** consistent colours, labels, titles and highlighted key points; accessibility for colour-blind viewers.
- **Medium:** slide, printed report or interactive dashboard.

Q7. Explain how dashboards and storytelling complement each other in Business Intelligence.

A dashboard provides a live, interactive view of KPIs for monitoring and exploration, but on its own it shows *what* is happening and leaves interpretation to the user. Storytelling adds the narrative: it picks the important findings from the dashboard, explains why they matter and recommends action. Dashboards supply the evidence and up-to-date data; storytelling supplies the meaning and the call to action. Together they support both everyday monitoring and strategic decisions.

Q8. What challenges may arise while communicating analytical insights to non-technical stakeholders?

- Technical jargon (p-values, RMSE, clustering) that the audience does not understand.
- Too much information or too many complex charts, which hides the main message.
- Different priorities and levels of data literacy among stakeholders.
- Lack of trust in data or resistance to results that contradict their opinions.
- Uncertainty and limitations of models that are hard to explain simply.
- Not linking the results to business impact, so no action is taken.
- Misinterpretation of correlation as causation.

Q9. Give two real-world examples where data storytelling has influenced business or policy decisions.

- **John Snow's cholera map (London, 1854):** by mapping cholera deaths around the Broad Street water pump, Snow showed that cases clustered around that pump. This evidence persuaded local authorities to remove the pump handle and became a landmark in public-health and epidemiology policy.
- **Florence Nightingale's 'rose' (coxcomb) diagrams:** she used charts showing that most British soldiers in the Crimean War died from preventable diseases rather than battle wounds. The visual story helped convince officials to improve sanitation and hospital reforms in the army.

(Both are well-known historical cases; you may also add a business example, such as a company reducing discounts after a profitability analysis like the one in this experiment.)

Q10. How can effective data storytelling improve strategic planning and organizational performance?

- Turns analysis into clear, prioritized actions that leadership can approve quickly.
- Aligns teams around the same facts and goals (a shared understanding of performance).

- Reveals opportunities and risks early, supporting better forecasting and resource allocation.
- Encourages a data-driven culture, as employees see how data leads to results.
- Improves communication between analysts and management, so insights are implemented rather than ignored.
- Allows progress to be tracked with KPIs, leading to continuous improvement and higher organizational performance.

Result: A complete data story with KPIs, visualizations, insights and recommendations was created for the selected dataset.